

Components used:

- 1] Atmega328p-AU
- 2] MPU6050
- 3] TMF8820
- 4] Two motor driver ic's (L293D)
- 5] Servo motor
- 6] 7.4v to 5v Voltage Regulator
- 7] 5v to 3.v Voltage Regulator
- 8] 4 dc motors

1] 3.3V regulator current

Each sensor: 10-20 mA

Two sensors: 40mA

Safety current sent: 100mA

2] Power dissipation in regulator

V_{in}: 5V

V_{out}: 3.3V

V_{drop}: 1.7v and P: 58mW - safe

3] Battery

7.4V Li-ion battery

buck converter: 5v and 3A

4] Atmega328P-AU

I_{mcu}: 20mA

5] MPU6050

I: 3.9mA

6] TMF8820

I: 30 mA

7] Servo Motor

I= 0.8A

8] DC Motors

I: 400mA

four motors: 1.6A

Total Current : (20 + 3.9 + 30)mA + 1.6A + 0.8A
which is 2.4539A.

9] L293D

V_{motor}: 4.6V

